Temperature and Migration in China

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1 Introduction

Migration is one of the most important channels through which households adjust to economic, social, and environmental shocks. While a growing literature has explored climate's impact on migration, labor supply, and productivity, less attention has been paid to how extreme temperature has shaped countries in the process of mass urbanization, like China.

How do extreme temperature events causally affect migration flows? China presents a compelling setting to study this question. There is substantial climatic heterogeneity among regions in China, from icy Harbin known for its ice sculpture festival to furnace cities like Chongqing and Wuhan, named so for their blazing summers. Migration is a social and structural focus in China as well, with the *hukou* identification system, which tracks and sometimes limits migration within China. Urbanization has also been very important in China as it has developed over the past decades to become the world's second-largest economy.

This paper uses a two-way fixed-effects regression to estimate the causal impact of extreme temperature on migration rates between provinces in China. I find that one more day where the average temperature rises above 90° F raises the inter-provincial migration rate in China by .19 percentage points. I compare this baseline result to additional analyses using covariates and robustness checks, including lagged temperature exposure. These findings have implications for the reform of the *hukou* system limiting migration, as well as the climate infrastructure and investment ecosystem in China.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature on climate shocks, migration, and adaptation. Section 3 describes the data used for migration and temperature. Section 4 outlines the empirical strategy. Section 5 presents the main results and heterogeneity analyses. Section 6 provides robustness checks related to alternative fixed effects, alternative clustering, and lagged temperature exposure. Section 7 concludes.

2 Background

Chen, Oliva Zhang (2022) examine the impact of air pollution on migration in China from 1996 to 2010 using census data. They found that a five-year increase in air pollution was enough to reduce migration inflows into that county by 50% and outflows by 5%, after accounting for national and county-level baseline migration flows. Interestingly, the authors also note that this effect is especially pronounced for early professionals. They use medium-run random variation in air pollution stemming from 5-year fluctuations in the strength of thermal inversions in a given county to address the likely correlation between pollution and economic activity, and thus pollution and migration, even after using county fixed effects.

Xia, Qingchun, and Baptista (2022) use census data from 2009 to 2011 to evaluate what characteristics best predict internal migration in China. They identify and assess fifteen economic, demographic, environmental, locational, and land-related factors, with the specific environmental factors that they assess being related to food production: river density, accumulated temperature and soil nutrients. They found

generally that migration happened from inland to the coast, and from rural to urban areas, with heterogeneous results for which characteristics best predicted inflows and outflows. Overall, GDP, population density, and labor force proportion drove internal migration.

Gray, Hopping, and Mueller (2020) examine climatic effects on internal migration from 1989 to 2011 with China Migrants Dynamic Survey data, methodologically using fixed effects regressions to measure climatic effects on migration and attrition and controlling for spatial and temporal confounders. Ultimately, they find that the effect of temperature on permanent migration transitioned from positive to negative over the course of the study, after taking into account trends over time and interactions between time and demographic controls. They speculate that this relates strongly to an increase in adaptive tools over time, making households more resistant to climate shocks such as extreme temperature.

Wang et al. (2022) use 2009 to 2018 data from the China Migrant Dynamic Survey to explore the relationship between temperature and permanent migration patterns of older people. They found that extreme cold was a driving force for migration of older people, but not extreme heat, and also that older people were more likely to move to places that were relatively warmer than their origin, not necessarily places that were not cold at all.

The literature clearly shows that the connection between internal migration in China and climatic effects has been probed. My paper adds to this body of literature by providing another estimate for the causal impact of temperature exposure on migration with different data (individual-level CPFS data aggregated to the provincial level) and also by providing robustness checks such as lagged temperature exposure to more confidently report results. The individual-level covariates also allow for heterogeneity analyses in my paper that further color my results.

3 Data

I combine data from the China Panel Family Studies (CPFS) from 2010-2018 with province-level measures of temperature from the Global Historical Climatology Network (GHCN) using GADM (Global Administrative Areas) data. The CPFS provides detailed individual-level demographic information, allowing me to construct variables measuring migration based on the changing location of a respondent over time. I construct a balanced panel of province–year observations by averaging individual-level outcomes within each province and survey year. Observations with fewer than 30 underlying individuals are dropped to reduce sampling noise, leaving a province–year panel with sufficient variation for fixed-effects estimation. Additionally, I construct three variables for temperature exposure: the count of days per year in which the average temperature exceeds 90°F, lies between 80–90°F, or falls below 40°F.

The following table shows descriptive statistics for the variables used in this paper, aggregated at the province-year level.

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
migration_rate	129	0	0.1	0.2	0.0	0.0	1.0
above90	30	0	7.8	11.7	0.0	5.0	93.0
day8090	100	0	77.1	43.1	1.0	74.5	193.0
below40	102	0	92.3	66.1	0.0	91.0	220.0
avg_income	154	0	21 009.0	13 994.0	2558.6	15 740.8	66 607.0
avg_age	139	9	43.9	7.5	30.3	43.9	75.0
pct_male	123	0	0.6	0.2	0.5	0.5	1.0
pct_urban	152	0	0.5	0.2	0.1	0.5	1.0

4 Methods

To estimate the effect of temperature on migration, I employ a two-way fixed effects regression with the following form:

$$Migration_{pt} = \beta_1 Above90 + \beta_2 Day8090 + \beta_3 Below40 + \gamma_p + \delta_t + \epsilon_{pt} \tag{1}$$

where γ_p and δ_t denote province and year fixed effects. Standard errors are clustered at the province level. In words, this regression looks at the effect of each temperature-exposure bin on the migration rate, after takig into account the time-invariant characteristics by province and trends happening over all provinces by year.

The variation described by this TWFE regression comes from viewing short-run fluctuations in temperature within provinces as quasi-random shocks. An analysis of lagged temperature exposures to motivate this interpretation is contained in Section 5. Conceptually, I find bouts of extreme temperature to be plausibly exogenous due to the incorporation of fixed effects which absorb time-invariant climatic differences already.

I estimate a second TWFE regression, adding in demographic covariates averaged to the provincial level. The form is as follows:

$$\begin{aligned} Migration_{pt} = \beta_1 Above90 + \beta_2 Day8090 + \beta_3 Below40 \\ + \beta_4 Income + \beta_5 Age + \beta_6 Male + \beta_7 Urban + \gamma_p + \delta_t + \epsilon_{pt} \end{aligned} \tag{2}$$

where standard errors are again clustered at the province level. In words, this is the same regression as before but includes demographic covariates in case any of them drive omitted variable bias in the first regression.

Finally, I conduct heterogeneity analyses by running the initial baseline TWFE regression, split by demographic group (i.e. male vs female, urban vs rural, and then by income terciles). The purpose of this is to further probe whether any demographic characteristics drive changes in migration rate, rather than the temperature exposure.

5 Results

Table 1 reports the baseline TWFE with province and year fixed effects.

Table 1: Temperature Effects on Migration

	(1)
Days > 90 F	0.0019*
	(1×10^{-3})
Days 80-90 F	0.0004
	(4×10^{-4})
Days < 40 F	−0.0003
	(6×10^{-4})
Num.Obs.	153
R2	0.847
R2 Adj.	0.803

* p <0.1, ** p <0.05, *** p <0.01
Standard errors clustered by province in parentheses.

The baseline TWFE regression yields a positive and statistically marginal effect of extreme heat on inter-provincial migration. The estimated coefficient on days above 90°F is 0.0019 with province-clustered standard errors suggesting this is significant at $\alpha = .10$ threshold. This magnitude implies that experiencing one more hot day raises the migration rate by approximately 0.19 percentage points. In contrast, the estimated effects of moderate heat and extreme cold are even smaller and statistically indistinguishable from zero. These coefficients suggest that only extreme heat affects migration decisions, not moderate heat nor extreme cold.

A visualizaton of this regression using residualized variables (to represent fixed effects) for extreme heat only is contained in Figure 1, and a coefficient plot for all temperature bins is shown in Figure 2 at the 90% confidence level.

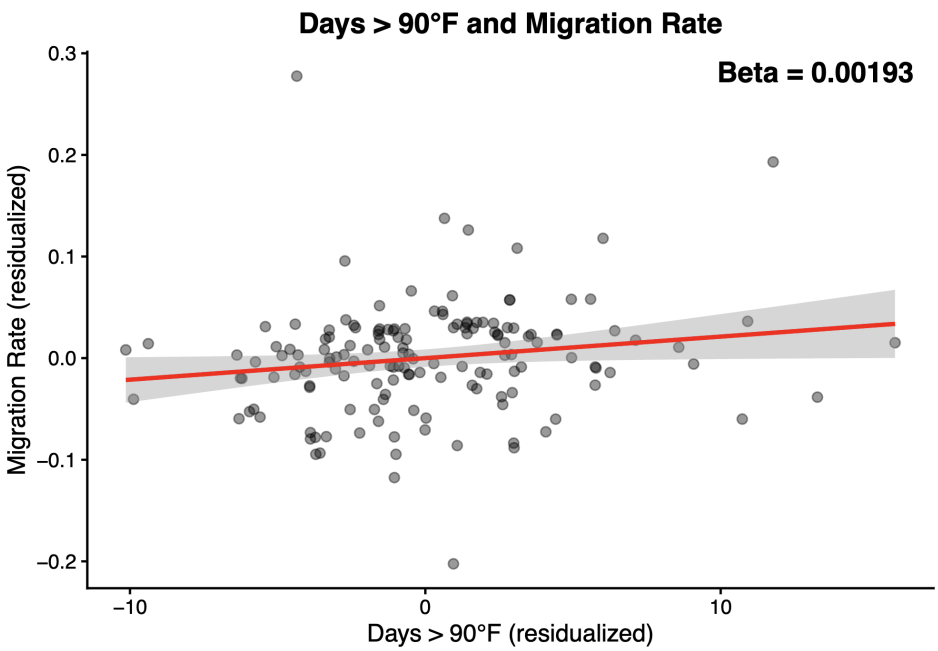

Fig. 1 Residualized plot of migration rate and extreme heat days

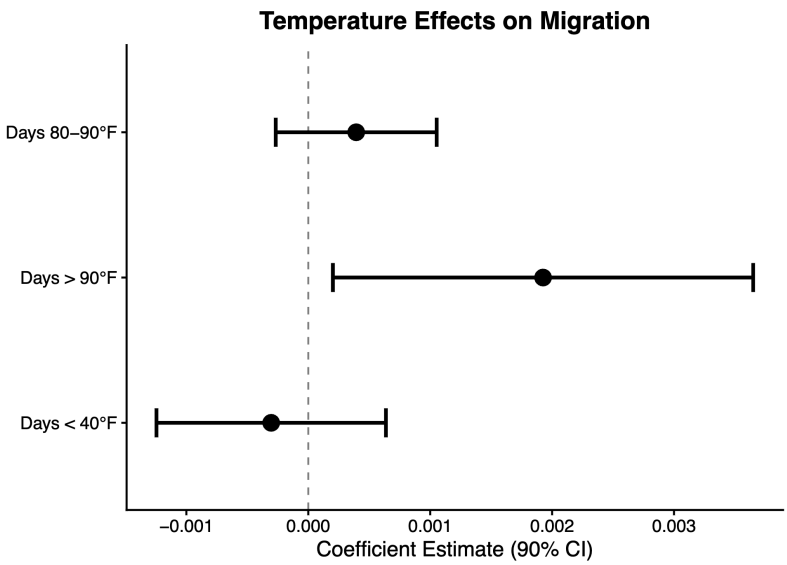

Fig. 2 Coefficient plot of migration rate and temperature exposure

Table 2 adds province-year covariates (income, age, gender, and urban status) to the baseline specification. In this model, the coefficient on extreme heat days remains positive and of a similar magnitude, but is no longer significant at the 10 percent level. This suggests that the positive relationship between extreme heat and migration is robust in sign (positive) but not as strongly supported in terms of significance.

Table 2: Temperature Effects on Migration: Covariates

	(1)
Days > 90 F	0.0018
	(0.0011)
Days 80-90 F	0.0005
	(0.0006)
Days < 40 F	−0.0006
	(0.0007)
Average Income	0.0000
	(0.0000)
Average Age	−0.0013*
	(0.0008)
% Male	0.1348
	(0.1803)
% Urban	−0.0550
	(0.1308)
Num.Obs.	139
R2	0.684
R2 Adj.	0.568

* p <0.1, ** p <0.05, *** p <0.01
Standard errors clustered by province in parentheses.

The heterogeneity analyses I conduct split the baseline TWFE by gender (Table 3), urban status (Table 4), and income (Table 5). Table 3 shows a significant coefficient for extreme heat days for the male group (denoted by sample:TRUE) but not the female group, suggesting stronger male responsiveness to extreme temperature in migration. Table 4 shows that the rural group (denoted by sample:FALSE) has a significant coefficient for extreme heat days. This similarly suggests greater rural responsiveness, though the actual numeric difference is very small. There are no significant coefficients

in Table 5, which measures heterogeneity between income terciles, marked as high income, low income, and medium income. This suggests there are no statistically different migratory responses to extreme temperature between income level. Taken as a whole, these heterogeneity analyses suggest there is no one demographic driver of migration pushing the results that show up in Table 1.

Table 3: Temperature Effects on Migration: Heterogeneity by Gender

	sample: FALSE	sample: TRUE
Days > 90 F	7×10^{-4}	0.0011*
	(5×10^{-4})	(5×10^{-4})
Days 80-90 F	1×10^{-4}	0.0001
	(2×10^{-4})	(3×10^{-4})
Days < 40 F	0	−0.0002
	(2×10^{-4})	(3×10^{-4})
Num.Obs.	38 880	39 152
R2	0.666	0.679
R2 Adj.	0.666	0.679

* p <0.1, ** p <0.05, *** p <0.01
Standard errors clustered by province in parentheses.

Table 4: Temperature Effects on Migration: Heterogeneity by Urban/Rural

	sample: FALSE	sample: TRUE
Days > 90 F	9×10^{-4}*	8×10^{-4}
	(5×10^{-4})	(7×10^{-4})
Days 80-90 F	2×10^{-4}	1×10^{-4}
	(2×10^{-4})	(3×10^{-4})
Days < 40 F	-1×10^{-4}	-1×10^{-4}
	(2×10^{-4})	(3×10^{-4})
Num.Obs.	42 004	36 028
R2	0.642	0.702
R2 Adj.	0.642	0.702

* p <0.1, ** p <0.05, *** p <0.01
Standard errors clustered by province in parentheses.

Table 5: Temperature Effects on Migration: Heterogeneity by Income Level

	sample: High Income	sample: Low Income	sample: Medium Income
Days > 90°F	0.0012	8×10^{-4}	6×10^{-4}
	(0.0009)	(5×10^{-4})	(5×10^{-4})
Days 80-90°F	0.0000	2×10^{-4}	0
	(0.0004)	(3×10^{-4})	(2×10^{-4})
Days < 40°F	−0.0002	-1×10^{-4}	0
	(0.0004)	(2×10^{-4})	(2×10^{-4})
Num.Obs.	20 864	37 626	19 541
R2	0.688	0.644	0.677
R2 Adj.	0.687	0.644	0.676

* p <0.1, ** p <0.05, *** p <0.01
Standard errors clustered by province in parentheses. Sample split by income terciles.

6 Robustness checks

I conduct several robustness checks. First, I compare alternative fixed-effects specifications. I also compare alternative clustering specifications. Finally, I conduct a

placebo test by using lagged temperature. The robustness checks generally show that the results in Table 1 are stable: extreme heat is the only variable of temperature exposure that drives migration, and that there is a positive causal effect of extreme heat on migration.

Table 6 shows that extreme heat days have significant coefficients when using fixed effects for province and year, or province only. The coefficient on extreme heat is also the same for three of the columns, falling to 0.0013 once province × year trend fixed-effects are used. This is consistent with the idea that some of the variation exploited in simpler fixed-effects are absorbed with province × year fixed effects. Importantly, this table shows that the sign on extreme heat is stable across alternative fixed effects.

Table 6: Robustness: Alternative Fixed Effects Specifications

	Province + Year FE	Province FE Only	Year FE Only	Province × Year Trend
Days > 90°F	0.0019*	0.0019*	0.0019	0.0013
	(1×10^{-3})	(0.0010)	(0.0012)	(0.0011)
Days 80-90°F	0.0004	0.0004	0.0004	0.0000
	(4×10^{-4})	(0.0004)	(0.0004)	(0.0005)
Days < 40°F	−0.0003	−0.0003	−0.0003	−0.0001
	(6×10^{-4})	(0.0006)	(0.0006)	(0.0007)
Num.Obs.	153	153	154	153
R2	0.847	0.847	0.875	0.916
R2 Adj.	0.803	0.803	0.839	0.862
R2 Within	0.032	0.530	0.839	0.016
R2 Within Adj.	0.007	0.494	0.800	−0.016
AIC	−392.0	−392.0	−394.0	−432.9
BIC	−286.0	−286.0	−284.7	−248.1
RMSE	0.05	0.05	0.05	0.04
Std.Errors	by: provcd	by: provcd	by: provcd	by: provcd
FE: year	X		X	X
FE: provcd	X	X		X

* p <0.1, ** p <0.05, *** p <0.01
Standard errors clustered by province.

Table 7 shows similar stability in the positive sign on extreme heat in causing migration rate. The coefficient is stable as well over different clustering specifications at .0019, which is significant when clustering at the province level and two-way clustering (both province and year). I mostly disregard the significant moderately-hot day coefficient in the two-way clustering, as it does not show up in any other regression. This robustness check continues to support that there is a positive effect on migration caused by extreme heat.

Table 7: Robustness: Alternative Clustering Structures

	Cluster: Province	Cluster: Year	Two-way Cluster
Days > 90°F	0.0019*	0.0019	0.0019*
	(1×10^{-3})	(0.0011)	(8×10^{-4})
Days 80-90°F	0.0004	0.0004	0.0004*
	(4×10^{-4})	(0.0003)	(2×10^{-4})
Days < 40°F	−0.0003	−0.0003	−0.0003
	(6×10^{-4})	(0.0003)	(3×10^{-4})
Num.Obs.	153	153	153
R2	0.847	0.847	0.847
R2 Adj.	0.803	0.803	0.803
R2 Within	0.032	0.032	0.032
R2 Within Adj.	0.007	0.007	0.007
AIC	−392.0	−392.0	−392.0
BIC	−286.0	−286.0	−286.0
RMSE	0.05	0.05	0.05
Std.Errors	by: provcd	by: year	by: provcd & year
FE: provcd	X	X	X
FE: year	X	X	X

* p <0.1, ** p <0.05, *** p <0.01
Testing sensitivity to clustering assumptions.

The placebo regression incorporates lagged temperature exposure comparing past and current temperate shocks. In running this robustness test, I find that the coefficients using current temperature shocks are mostly similar to the baseline TWFE regression. None of the coefficients on the current nor lagged temperature shocks are statistically significant, even at the α=.10 level. Due to the lack of statistically significant coefficients for the lagged analysis, this placebo test overall supports my empirical strategy. However, the modest effect I report in Table 1 may be influenced even by small lags, so this test may hint at an important limitation to consider in future work.

Table 8: PLACEBO TEST: Lagged Temperature Effects

	Current Temperature	PLACEBO: Lagged Temp	Both Current & Lagged
Days > 90°F (current)	0.0018		0.0019
	(0.0012)		(0.0013)
Days 80-90°F (current)	0.0001		−0.0003
	(0.0005)		(0.0005)
Days < 40°F (current)	−0.0009		−0.0010
	(0.0008)		(0.0008)
Days > 90°F (lagged)		0.0011	0.0013
		(0.0012)	(0.0011)
Days 80-90°F (lagged)		0.0007	0.0006
		(0.0005)	(0.0005)
Days < 40°F (lagged)		−0.0007	−0.0006
		(0.0005)	(0.0005)
Num.Obs.	126	126	126
R2	0.750	0.750	0.757
R2 Adj.	0.664	0.664	0.663
R2 Within	0.030	0.030	0.059
R2 Within Adj.	−0.001	−0.001	−0.004
AIC	−322.7	−322.6	−320.4
BIC	−229.1	−229.0	−218.3
RMSE	0.05	0.05	0.05
Std.Errors	by: provcd	by: provcd	by: provcd
FE: provcd	X	X	X
FE: year	X	X	X

* p <0.1, ** p <0.05, *** p <0.01
Lagged temperature should not predict current migration if relationship is causal.

7 Conclusion

The empirical results point to a consistent, statistically modest effect of extreme heat on inter-provincial migration in China. The baseline model without any additional controls finds that one more extreme heat day raises the migration rate by .19 percentage points. The addition of demographic covariates reduces this significance but preserves the point estimate. Heterogeneity, robustness, and placebo analyses largely preserve the sign of the extreme-heat coefficient but fail statistical significance.

What this may point to is the institutional context of migration in China. Structural restrictions, such as the *hukou* identification system, which is intended to limit migration especially to larger cities, may burden people with extensive application materials to officially change residences, in addition to the expected cost of housing and job relocation. This friction may mute a larger migration response expected from extreme-temperature. Against this backdrop, the fact that extreme heat still yields some detectable, albeit modest, effect on migration may suggest that climate stress is

beginning to register in migration behavior but has not yet overcome existing frictions for most households.

These results also interact with existing concerns in China over climate risk and water inequality. Regions that are both heat-exposed and water-stresesd may experience amplified incentives to move even in the face of institutional barriers like *hukou* applications. Future work could explore staggered (and qualitative) relaxation of *hukou* restrictions to analyze this question using minor policy shocks.

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